UNSTOP DIGITAL TWIN CHALLENGE 2026

GlucoTwin AI

Clinical Decision Support & Physiological Digital Twin Platform for Type 2 Diabetes

- Team: Karpagam Institute of Technology, Coimbatore, Tamil Nadu, India
- Domain: Healthcare & AI-driven Clinical Digital Twins
- Platform: Real-time CGM Telemetry, XGBoost Spike Forecasts, What-If Simulations

CLINICAL BACKGROUND

Executive Summary & Problem Statement

The Challenge in Type 2 Diabetes Management

- Clinical Burden: Type 2 Diabetes affects >500M individuals worldwide; postprandial glucose excursions (>10.0 mmol/L) drive vascular complications.
- Static EHR Limitation: Conventional clinical records provide only periodic snapshots (HbA1c quarterly) without continuous physiological context.
- Wearable Telemetry Disconnect: Modern Continuous Glucose Monitors (CGM) stream massive data, but clinicians lack predictive decision tools.
- The Opportunity: A versioned Physiological Digital Twin synthesizing static clinical baselines with real-time biometric telemetry.

SOLUTION ARCHITECTURE

Proposed Solution & Healthcare Use Case

Dual-Layer Digital Twin with Forward Risk Forecasting

- Dual-Layer Twin Architecture: Combines static EHR clinical history with a dynamic wearable sliding-window buffer.
- 2-Hour Spike Forecaster: Machine learning pipeline predicting postprandial hyperglycemia (>= 10.0 mmol/L / 180 mg/dL) 2 hours in advance.
- SHAP Feature Attribution: Statistical breakdown explaining exactly which physiological signals (CGM trend, HRV, activity) influence the forecast.
- Counterfactual Simulations: Clinicians model hypothetical meals and physical activity before prescribing interventions.

TECHNICAL ARCHITECTURE

System Architecture & Microservice Topology

Decoupled, Event-Driven Clinical Pipeline

- Presentation Layer: React 18 SPA (Vite, TypeScript, Tailwind CSS dark theme, Recharts telemetry charts).
- Application Core: Spring Boot 3.3.0 REST Gateway with Spring Security (ROLE_CLINICIAN) & Flyway database migrations.
- Event Streaming Bus: Redis 7 Streams (glucotwin:wearable-events) processing streaming CGM, heart rate, and HRV readings.
- Persistence Tier: PostgreSQL 16 storing patient EHR, versioned Digital Twin states, and immutable audit logs.
- Machine Learning Service: Python FastAPI container hosting calibrated XGBoost 2.0.3 and SHAP explainability engines.

DIGITAL TWIN ENGINE

Physiological Digital Twin State Machine

Dual-Layer Representation & Lifecycle Transitions

- Static EHR Layer: Age, biological sex, BMI, diabetes onset year, baseline HbA1c, fasting glucose.
- Dynamic Wearable Layer: Rolling CGM glucose window (6-12 points), heart rate, HRV autonomic tone, sleep, and step counts.
- Status Lifecycle:
- • INITIALISED: Static EHR loaded; awaiting continuous streaming sensor data.
- • ACTIVE: Live wearable telemetry received within the last 60 minutes.
- • STALE: Telemetry elapsed > 60 minutes; predictions automatically carry data-staleness flags.

MACHINE LEARNING

AI / Machine Learning Approach & SHAP

Calibrated XGBoost Model with Local Explanations

- Algorithm: Pre-trained XGBoost 2.0.3 binary classification model.
- Inference Objective: Probability of blood glucose exceeding 10.0 mmol/L within a 2-hour forward horizon.
- Outputs Delivered:
- • Point Probability: Calibrated risk probability (0.0% to 100.0%).
- • Categorical Risk Tier: LOW (<30%), MODERATE (30-60%), HIGH (60-85%), CRITICAL (>=85%).
- • 95% Confidence Interval: Empirical bounds evaluating prediction certainty.
- • SHAP TreeExplainer: Directional feature attributions (e.g., cgm_current increases risk by +32.0%).

CLINICAL SIMULATION

What-If Counterfactual Simulations

Hypothetical Scenario Modeling with Transactional Memory Isolation

- Clinical Purpose: Allows clinicians and researchers to test meal carbohydrate intake (0-500g), physical exercise levels, and medication adherence.
- Absolute Memory Isolation Guarantee:
- • Simulation executes exclusively against an in-memory ephemeral twin clone.
- • Observed Digital Twin state and physiological history are strictly IMMUTABLE.
- • Results are tagged with SIMULATED provenance and isolated from real patient EHR.
- Verified Invariant: Tests confirm baseline observed glucose remains identical before, during, and after simulation runs.

DECISION SUPPORT

Multi-Agent Clinical Insights Engine

Collaborative Agent Reasoning with Deterministic Fallbacks

- RiskEvidenceAgent: Synthesizes EHR baselines, dynamic telemetry, and ML spike probabilities.
- ClinicalKnowledgeAgent: Retrieves clinical guidance (ADA Standards of Care, glycemic variability guidelines).
- InsightExplanationAgent: Formulates structured medical narrative (LLM-enabled with automatic deterministic rule-based fallback).
- InsightVerificationAgent: Verifies numeric claims, asserts safety disclaimers, and records pipeline execution trace.

QUALITY ASSURANCE

Testing, Validation & Build Verification

100% Passing Automated Verification Across the Stack

- Frontend Suite: 74 / 74 Vitest component tests passing; strict TypeScript check (0 errors); production build clean (912 modules).
- Backend Suite: Spring Boot JUnit 5 and jqwik property tests passing (BUILD SUCCESSFUL); executable bootJar compiled.
- ML Service Suite: 29 / 29 Pytest unit tests passing (100% pass rate) covering model deserialization, SHAP attributions, and CI bounds.
- End-to-End Workflows: 7 / 7 automated workflow integration scenarios verified (telemetry ingestion, prediction, insights, simulation).
- Cohort Persistence: 15 independently versioned patient records verified in PostgreSQL.

GOVERNANCE & LIMITATIONS

Known Limitations & Clinical Disclaimer

Honest Operational Scope & Safety Governance

- Operational Scope: Designed exclusively for research and clinical decision support; not an FDA/CE-cleared autonomous diagnostic device.
- Hardware Emulation: High-throughput telemetry streaming uses Redis event streams rather than physical Bluetooth LE hardware pairings.
- Local LLM Mode: Uses deterministic rule-based medical reasoning by default when OPENAI_API_KEY is not configured in local development.
- Safety Governance: All screens display explicit clinical safety disclaimers and immutable provenance tags.

ROADMAP

Future Scope & Roadmap

Scaling from Decision Support to Closed-Loop Systems

- Hardware Integration: Direct Bluetooth Low Energy (BLE) drivers for Dexcom G7, Abbott FreeStyle Libre 3, and smart insulin pens.
- EHR Interoperability: Production FHIR R4 and HL7 SMART-on-FHIR connectors for hospital Electronic Medical Record systems (Epic, Cerner).
- Deep Learning Sequencers: Exploring LSTM and Transformer-based temporal models for continuous multi-step trajectory forecasting.
- Automated Insulin Delivery (AID): Exploring research sandbox integrations with open-source artificial pancreas algorithms.